OSP Project 1 Report
Single-Cycle Baseline Processor

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I. Supported Commands

Instruction Class	Supported Instructions
R-type	add, sub, and, or, xor, sll, srl, sra, slt, sltu
I-type	addi, andi, ori, xori, slli, srli, srai, slti, sltiu
Memory	lw, sw
Branches	beq, bne, blt, bge
Jump	jal

II. Implemented Modules

adder.sv, alu.sv, branch_unit.sv, control_unit.sv, data_mem.sv, imm_gen.sv, instr_mem.sv, instr_parser.sv, mux2.sv, pc.sv, regfile.sv, single_cycle_top.sv

testbench.sv

III. Known Limitations

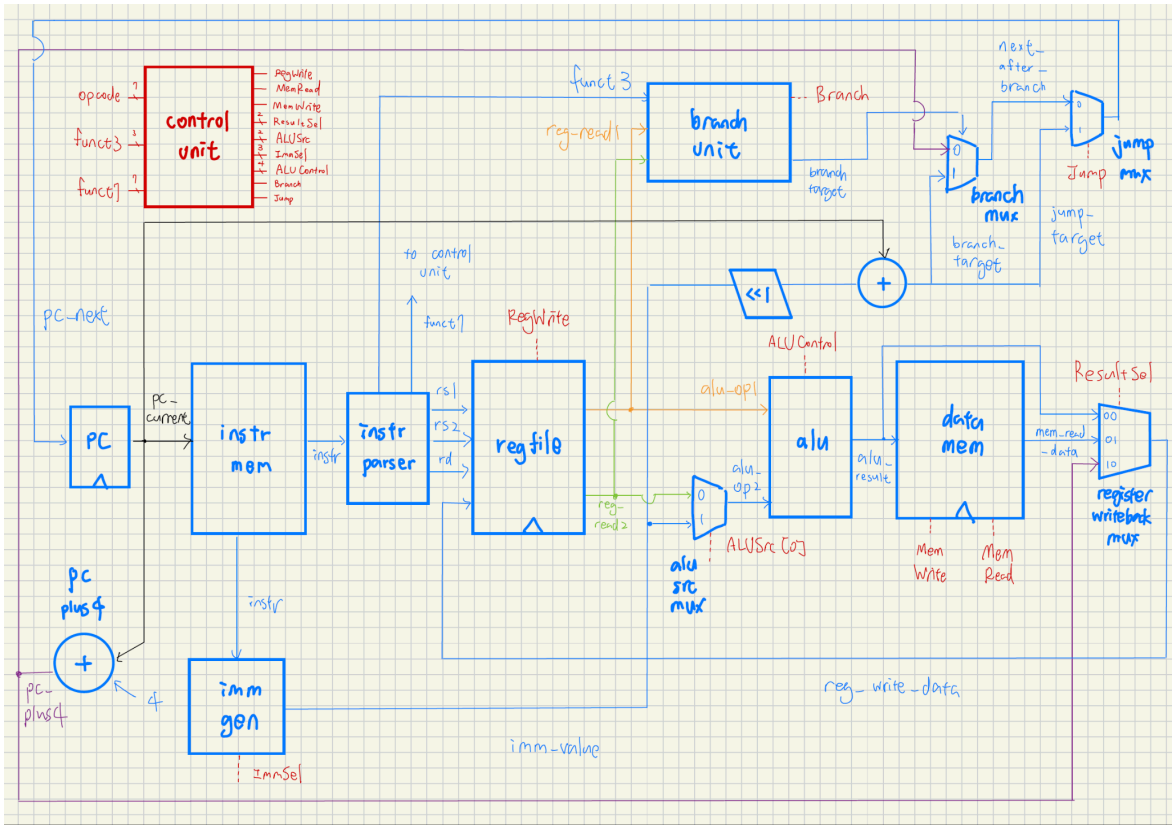
- This work only supports a subset of RISC-V instructions, especially U-type instructions, jalr are not supported.
- Although this project aims for synthesizable code, the design is unoptimized and will definitely destroy the timing closure of any implementation.

IV. Design Philosophy

- This work is a baseline design which requires further pipelining, PPA optimization to meet implementation demands.

V. Design

Block Diagram



Controller

1. RegWrite

Active for: R-type, I-type (including lw), jal

2. MemRead

Active for: lw

3. MemWrite

Active for: sw

4. ResultSel[1:0]

00: R-type, I-type (except for lw), sw (from alu)
01: lw (from dmem)
10: jal (from pc+4)
11: B-type, J-type (0)

5. ALUSrc[1:0]

00: R-type, B-type (from regfile)
01: I-type, sw (from imm)
10: jal (0)

6. ImmSel[2:0]

000: I-type
001: S-type
010: B-type
011: U-type
100: J-type

7. ALUControl[3:0]

0000 (0): add, addi, lw, sw
0001 (1): sub (Note: there is no such command as subi)
0010 ~ 1001 (2~9): the corresponding R-type and I-type instructions (1 for each):
0010: and
0011: or
0100: xor
0101: sll
0110: srl
0111: sra
1000: slt
1001: sltu
1010 (10): B-type, jal (0)

8. Branch

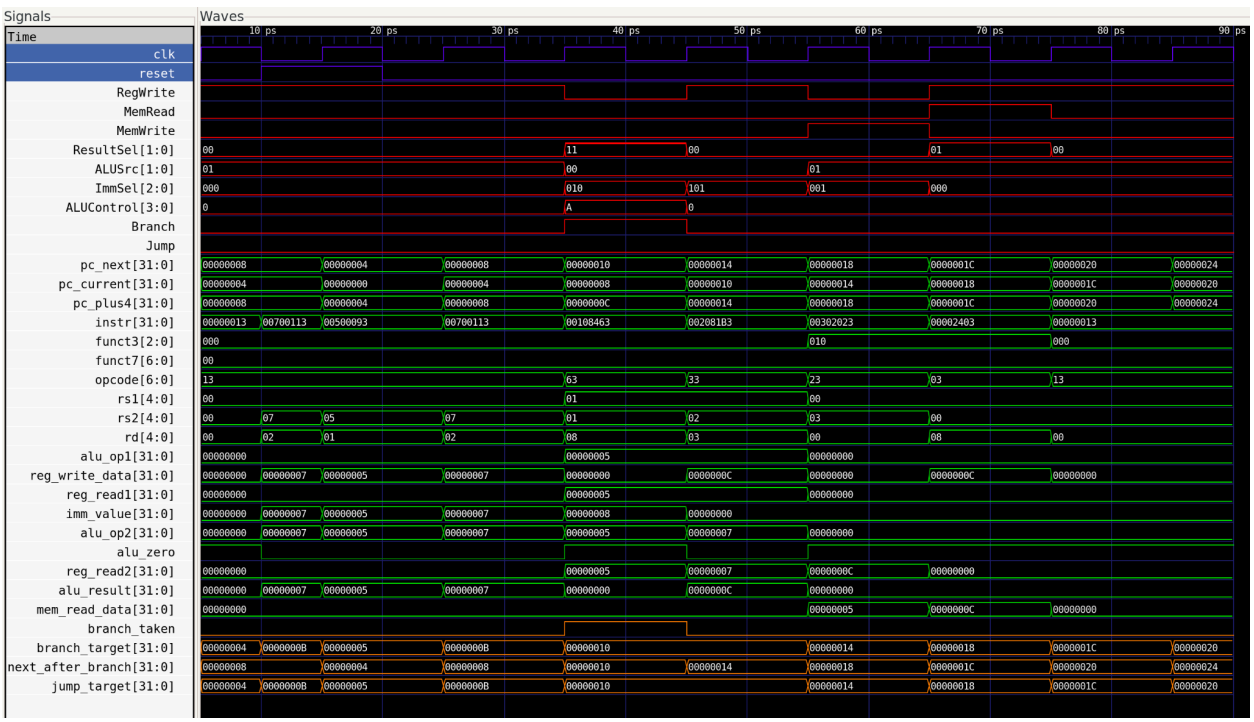
Active for B-type

9. Jump

Active for jal

VI. Verification

```
00500093 // addi x1(ra), x0(zero), 5
00700113 // addi x2(sp), x0(zero), 7
00108463 // beq x1(ra), x1(ra), 8
06300193 // addi x3(gp), x0(zero), 99
002081b3 // add x3(gp), x1(ra), x2(sp)
00302023 // sw x3(gp), x0(zero)(0)
00002403 // sw x0(zero), x0(zero), 0
00000013 // addi x0(zero), x0(zero), 0
```




```
root@aa410c9c2161:/work# ./obj_dir/Vtestbench
          5
          7
         12
         12
          0
- testbench.sv:52: Verilog $finish
```